

Set 4 exercises: Arrays

Exercise 1: (Lecture)

Write an algorithm that determines, from a list of 10 integers, the number of elements less than a given value **Val** (in two versions, with and without using vectors).

Exercise 2: (Lecture)

Given a vector of integers **A** of size **n** (**n < 100**), write an algorithm that:

1. Reads the values of vector **A** using two different methods.
2. Displays the **sum, maximum, and average** of the elements of **A**.
3. Divides by 2 **all even elements**, then doubles all elements at **odd positions**.
4. Copies the elements of **A** into another vector **B** of the same size.
5. Copies the **odd elements** of **A** into another vector **C** and then displays its **minimum**.
6. Performs a **right rotation** of the elements of **A**.
7. Performs a **left rotation** of the elements of **B**.
8. Constructs and displays a vector **D** such that:
 $D[n-1] = A[0] + B[0], D[n-2] = A[1] + B[1], \dots, D[0] = A[n-1] + B[n-1]$.

Exercise 3: (Lecture)

Let **val** be an integer value, and let **T** be a vector of **n** integer values ($1 \leq n \leq 50$). Write an algorithm that:

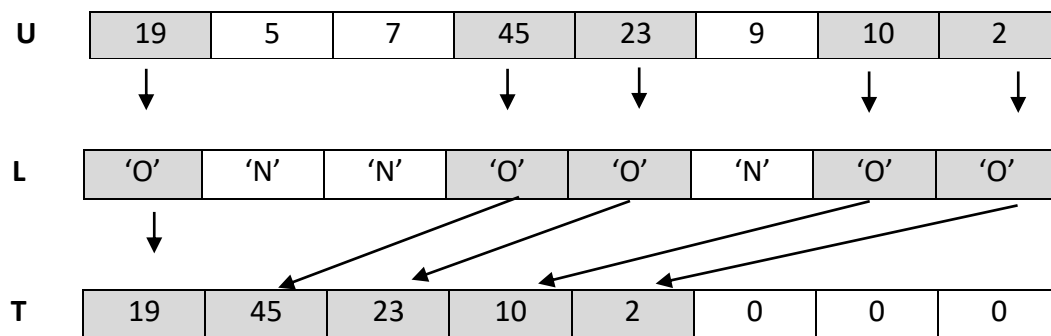
1. Reads **n, val**, and the elements of vector **T**.
2. **Checks** whether the value **val** exists or not in vector **T** using the **counter** method and the **flag** method.

Exercise 4: (Lecture)

Write an algorithm that:

1. Reads a vector **U** of size **n** ($0 < n \leq 100$) of integers, and a character array **L** of the same size containing only the characters '**O**' and '**N**'.
2. Constructs a new vector **T** of the same dimension, which will contain the elements of vector **U** that coincide with the value '**O**' in vector **L**. If there are any empty slots, they will be filled with zeros.
3. Displays vector **T**.

Example:



Exercise 5 : (Lecture)

Let **T** be an array of integers of size $n \leq 100$, containing both zero and non-zero values.

Write an algorithm that places the zero values at the beginning of the array **T** and the non-zero values at the end.

Example: T

1	0	-3	5	-9	0	-6
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Then:

T

0	0	1	-3	5	-9	-6
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Exercise 6: (Lecture)

Let consider a vector **T** of **n** integer values ($2 \leq n \leq 50$).

Write an algorithm that:

1. Reads the elements of the vector **T**.
2. Indicates **whether** the elements of the vector **form a regularly decreasing sequence** or not.

Note: A sequence is considered regular if: $T[0] - T[1] = T[1] - T[2] = \dots = T[n-2] - T[n-1]$.

Example:

T

15	12	9	6	3
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 The elements of T form a regularly decreasing sequence

Exercise 7 : (Lecture)

Write an algorithm that allows you to:

1. Read a vector of integers T of size ($n \leq 10$)
2. Read an integer **p** which represents a position in **T**
3. Read an integer **val**
4. Insert, if possible, the value **val** at position p in array **T**
5. Delete the first occurrence of **val** in the array **T**
6. Display the array **T**

Exercise 8

Write an algorithm that inputs two arrays of integers T1 and T2 with respective sizes n, m (where $n \leq 100$, $m \leq 50$), and calculates the **smurf** of the two arrays. To calculate the **smurf**, you need to multiply each element of T1 by each element of T2 and then sum the results.

For example, if we have:

T1

4	8	7	12
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T2

3	6
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The smurf will be :

$$3 \cdot 4 + 3 \cdot 8 + 3 \cdot 7 + 3 \cdot 12 + 6 \cdot 4 + 6 \cdot 8 + 6 \cdot 7 + 6 \cdot 12 = 279$$

Exercise 9

Let consider two vectors **T1** and **T2** of integers, with respective sizes **n1** and **n2** ($n1 \leq 50$, $n2 \leq 100$), assumed to be already sorted in ascending order of their values. Write an algorithm that constructs a sorted array T3 in ascending order by merging the arrays **T1** and **T2**.

Example: T1

-9	0	5	10	13
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T2

-3	-2	0	4	7	8
----	----	---	---	---	---

Then:

T3

-9	-3	-2	0	0	4	5	7	8	10	13
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